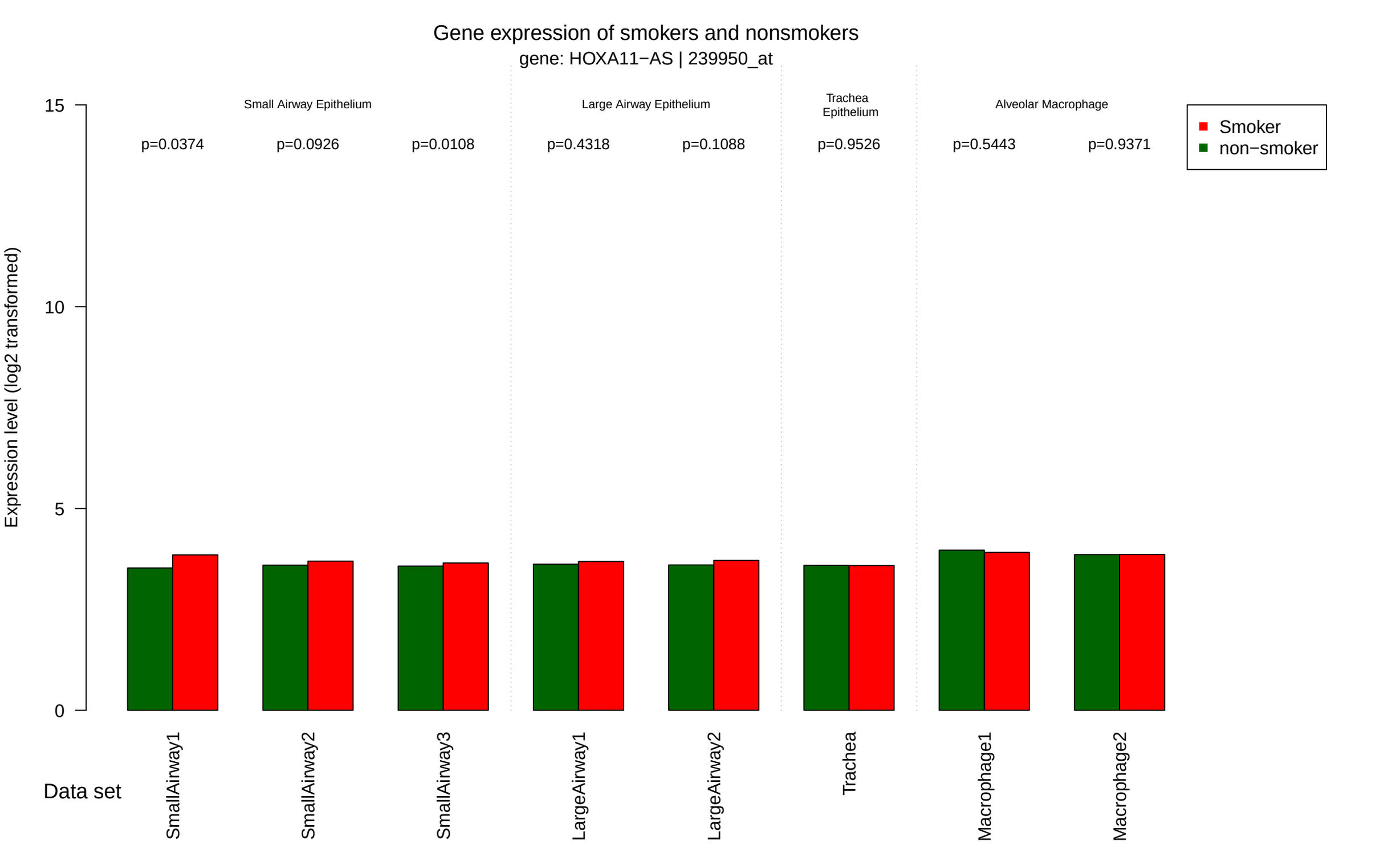
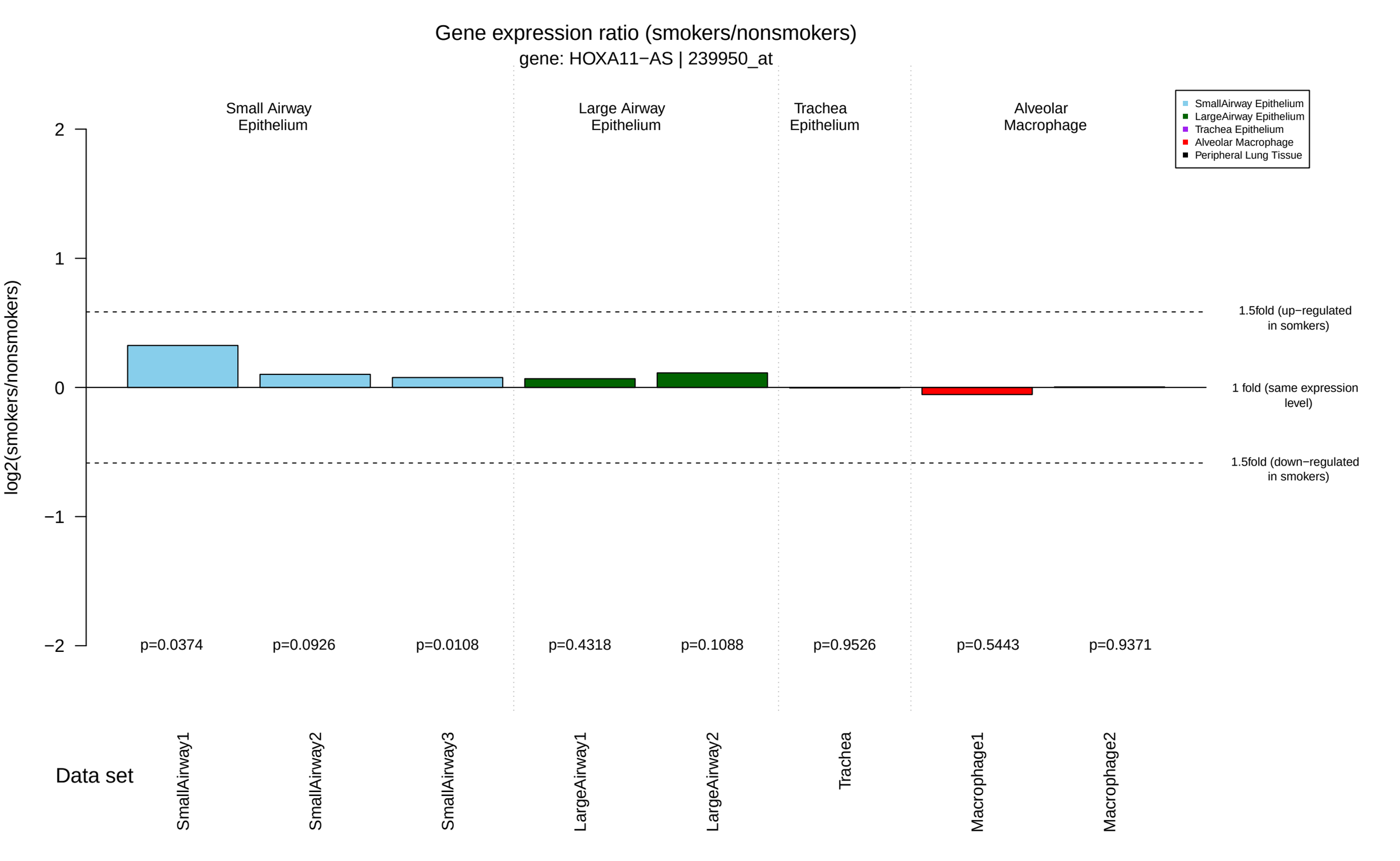
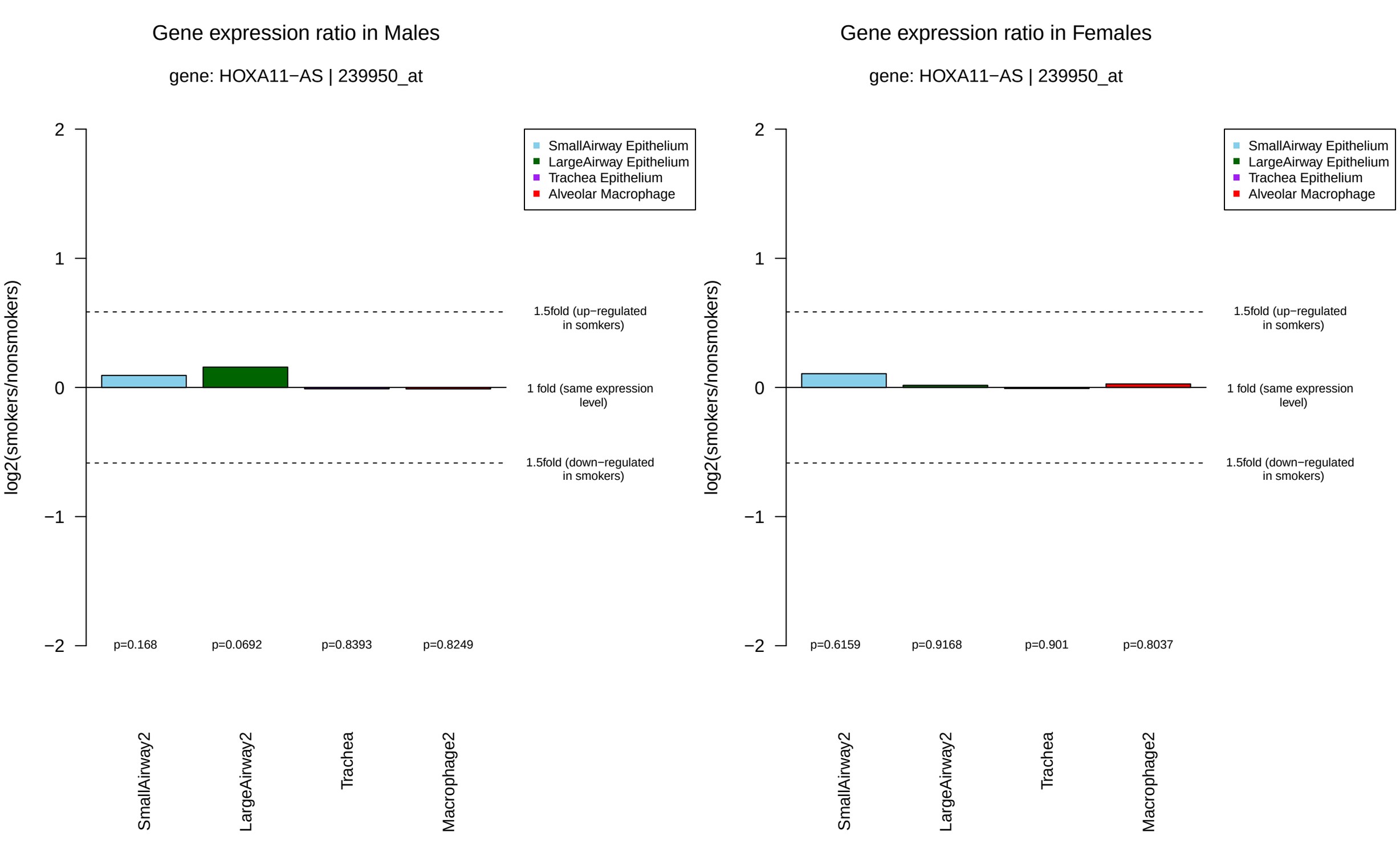
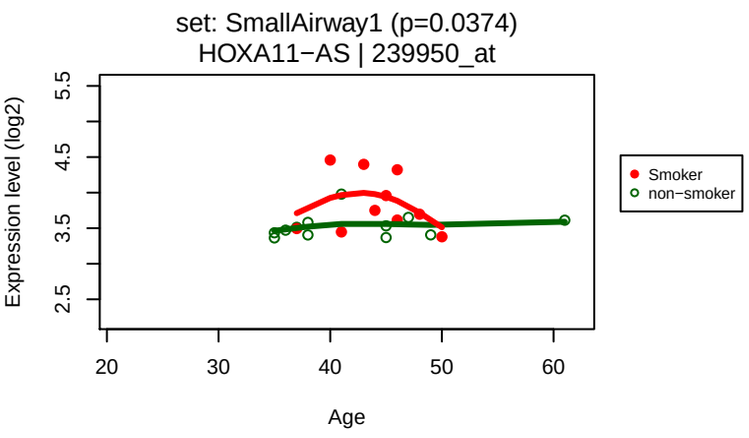
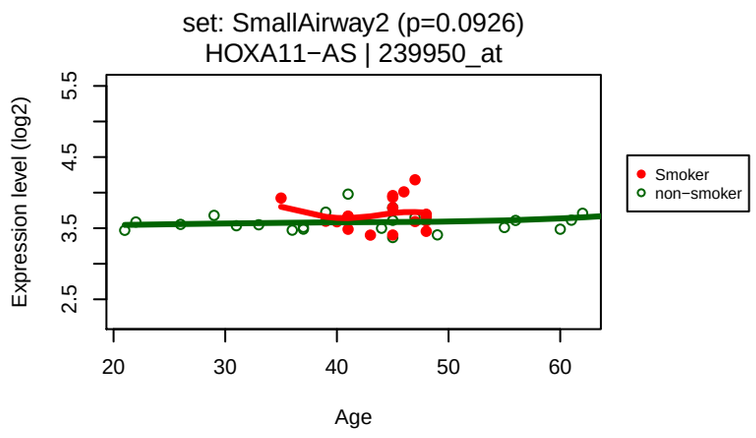
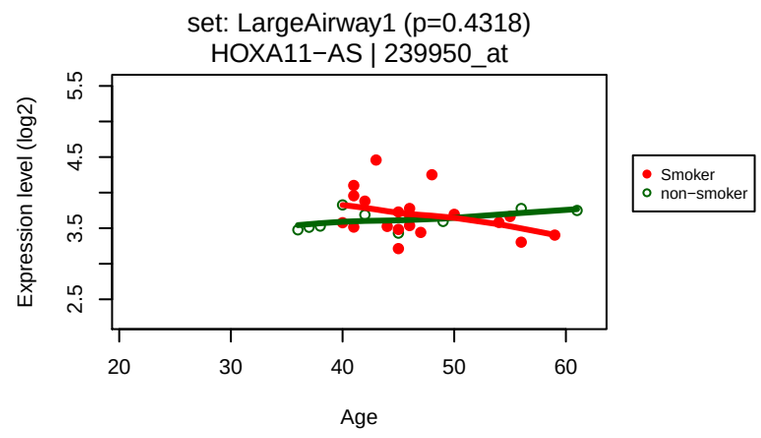

Gene expression of smokers and nonsmokers
gene: HOXA11−AS | 239950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0374
p=0.0926
p=0.0108
p=0.4318
p=0.1088
p=0.9526
p=0.5443
p=0.9371
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

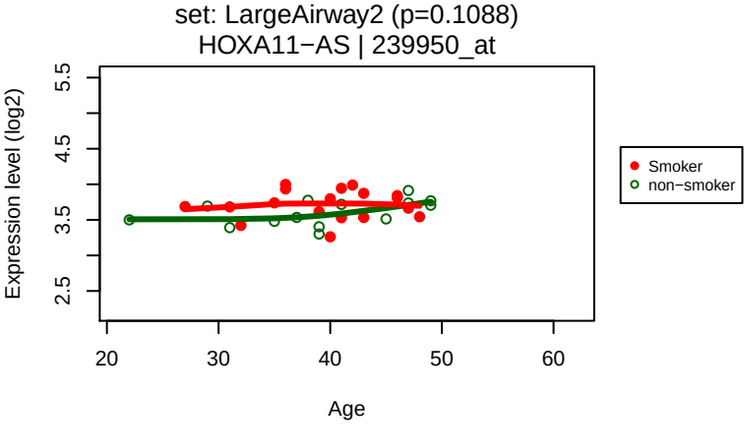

Gene expression ratio (smokers/nonsmokers)
gene: HOXA11−AS | 239950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0374
p=0.0926
p=0.0108
p=0.4318
p=0.1088
p=0.9526
p=0.5443
p=0.9371
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

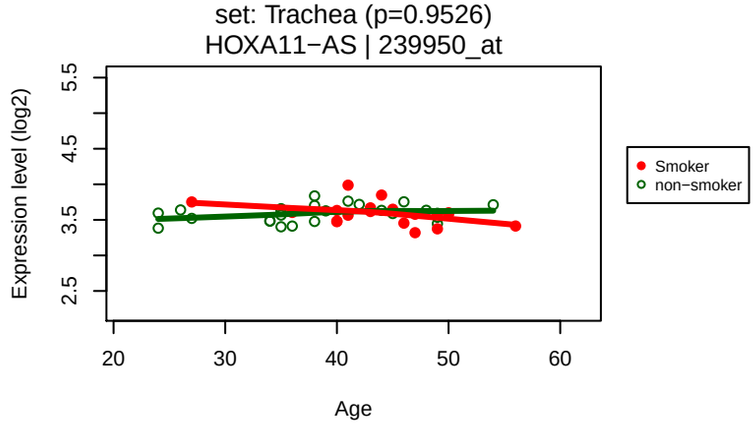

Gene expression ratio in Males
gene: HOXA11−AS | 239950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.168
p=0.0692
p=0.8393
p=0.8249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA11−AS | 239950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6159
p=0.9168
p=0.901
p=0.8037
SmallAirway2
LargeAirway2
Trachea
Macrophage2

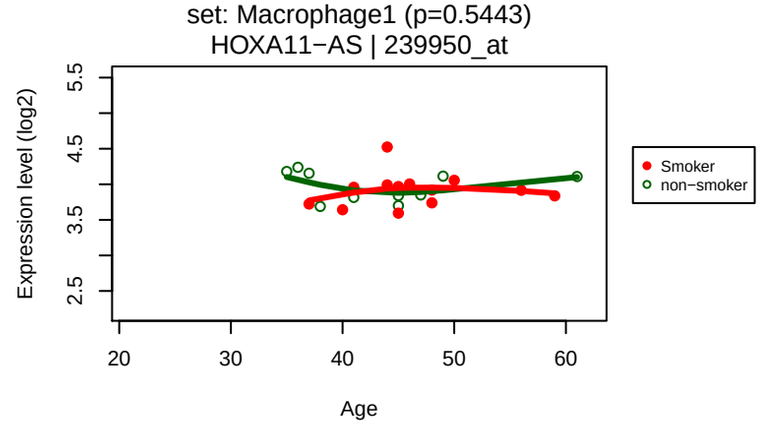

set: SmallAirway1 (p=0.0374)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker

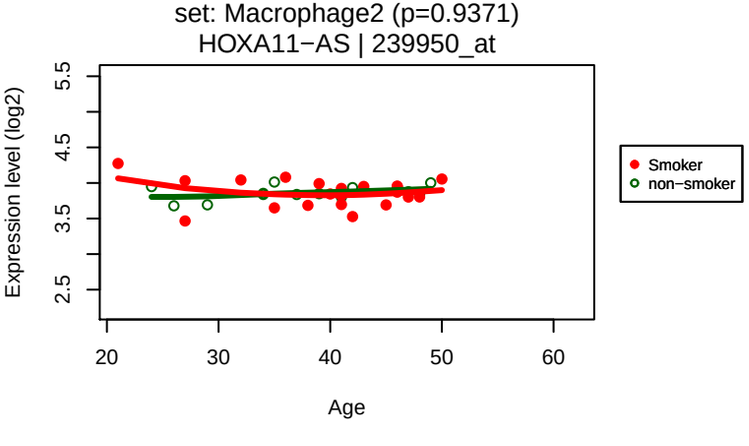

set: SmallAirway2 (p=0.0926)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker

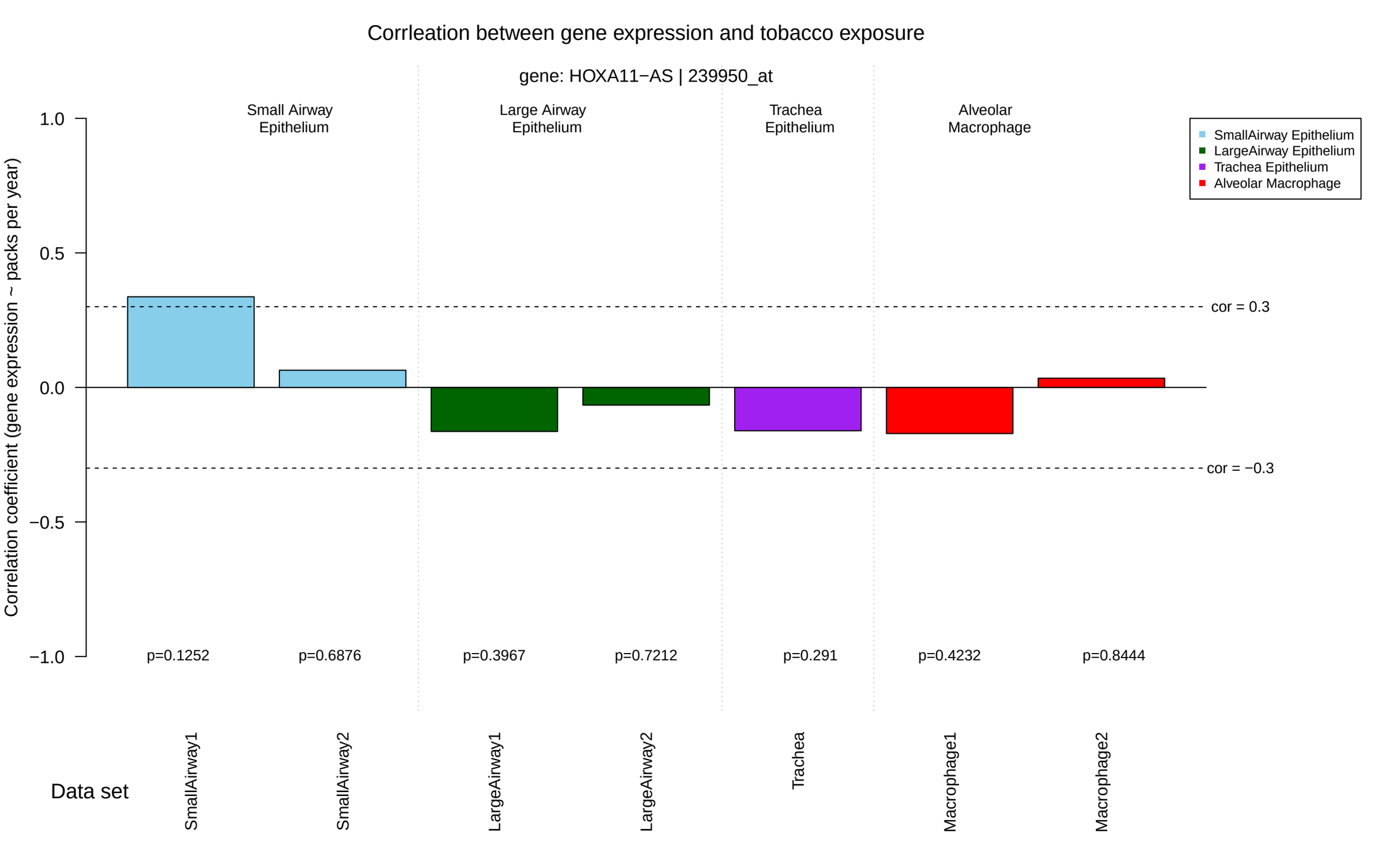

set: LargeAirway1 (p=0.4318)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1088)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9526)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5443)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9371)
HOXA11−AS | 239950_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA11−AS | 239950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1252
p=0.6876
p=0.3967
p=0.7212
p=0.291
p=0.4232
p=0.8444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2